

UNIVERSITY OF THESSALY
SCHOOL OF ENGINEERING
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Investigation of Machine Vibration Analysis System Design

Diploma Thesis

Vasileios Dimitriou

Supervisor: Karakonstantis Georgios

Month 2022



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ΠΑΝΕΠΙΣΤΗΜΙΟ ΘΕΣΣΑΛΙΑΣ

ΠΟΛΥΤΕΧΝΙΚΗ ΣΧΟΛΗ

ΤΜΗΜΑ ΗΛΕΚΤΡΟΛΟΓΩΝ ΜΗΧΑΝΙΚΩΝ ΚΑΙ ΜΗΧΑΝΙΚΩΝ ΥΠΟΛΟΓΙΣΤΩΝ

**Διερεύνηση Σχεδίασης Συστημάτων Ανάλυσης Δονήσεων
Μηχανών**

Διπλωματική Εργασία

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Diploma Thesis

Investigation of Machine Vibration Analysis System Design

Vasileios Dimitriou

Abstract

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term, term, . . . , term

Διπλωματική Εργασία

Διερεύνηση Σχεδίασης Συστημάτων Ανάλυσης Δονήσεων Μηχανών

Βασίλειος Δημητρίου

Abstract

Η περίληψη περιλαμβάνει την επιστημονική περιοχή και το σκοπό - αντικείμενο της εργασίας, τη μεθοδολογία, τα κύρια βήματα που ακολουθήθηκαν και τέλος τα κύρια αποτελέσματα. Η συνολική έκταση της περίληψης θα είναι μέχρι μία σελίδα.

Διπλωματική εργασία γραμμένη στα ελληνικά, θα πρέπει να περιλαμβάνει ελληνική και αγγλική περίληψη.

Λέξεις-κλειδιά:

όρος, όρος, . . . , όρος

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Abbreviations

βλπ	βλέπε
κ.λπ.	και λοιπά
κ.ο.κ	και ούτω καθεξής
TEI	Τεχνολογικό Εκπαιδευτικό Ίδρυμα
BPF	Band Pass Filter

Chapter 1

Introduction

Motivation and description. Points:

- Importance of maintenance
- Predictive maintenance
- Object of my thesis
- Contribution of my thesis

Εδώ αυτή κάνουμε μια γενική περιγραφή του χώρου εφαρμογής της διπλωματικής. Αναφέρουμε τα χαρακτηριστικά του χώρου και καταλήγουμε στα γενικότερα προβλήματα που αντιμετωπίζει ο χώρος. Η συζήτηση των προβλημάτων θα πρέπει να προϋποθέτει τον αναγνώστη για το τι θα προσπαθήσει να αντιμετωπίσει η διπλωματική, χωρίς ακόμα να αναφερόμαστε συγκεκριμένα στο αντικείμενο της διπλωματικής.

1.1 Object of the Diploma Thesis

Elaboration on a system that predicts the failure of mechanical components based on historical data. This thesis focuses on resolving the issue of unplanned shutdowns due to corrective maintenance of machines that have bearings. Εδώ αναφερόμαστε συγκεκριμένα στο τί θα κάνει η διπλωματική. Αναφέρουμε λεπτομερώς α) τα προβλήματα που θα λύσει (και που ήδη έχουν περιγραφεί γενικά στην προηγούμενη ενότητα), και β) πώς σκοπεύει να τα λύσει. Είναι σημαντικό κάποιος που θα διαβάσει την ενότητα αυτή να καταλάβει σε σημαντικό βαθμό τον σκοπό της διπλωματικής σας και τις τεχνικές δυσκολίες της, χωρίς

να είναι αναγκαίο να δει όλα τα άλλα κεφάλαια. Η ενότητα αυτή θέλει πολύ προσοχή και καλύτερα να τη γράψετε αφού έχετε γράψει όλα τα υπόλοιπα κεφάλαια.

1.1.1 The Concept of Predictive Maintenance

Getting a good grasp on the upcoming required maintenance activities and being aware of what is of utmost importance is vital for the operation of a plant. A grasp on Periodical status checks of different machines is crucial for the maintenance engineers, since this could yield a clear understanding of what priority is at that instance of time. Therefore the prediction of an upcoming machine failure would constitute an invaluable tool.

- Good condition readings
- Periodical readings
- Readings analysis
- online board

1.1.2 Contribution

Εδώ παραθέτουμε αριθμητικά συγκεκριμένες ενέργειες/λύσεις/μεθοδολογίες που παρουσιάζει η διπλωματική και λύνουν τα προβλήματα που υποσχθήκαμε στην προηγούμενη ενότητα ότι θα λύσει η διπλωματική. Συνήθως η υποενότητα αυτή έχει την παρακάτω μορφή:

Η συνεισφορά της διπλωματικής συνοψίζεται ως εξής:

1. Μελετήθηκαν συστήματα κ.λ.π.
2. Υλοποιήθηκαν τρεις αλγόριθμοι υπολογισμού κ.λ.π.
3. Αξιολογήθηκε η επίδοση των αλγορίθμων και βρέθηκε ότι κ.λ.π.
4. Ενσωματώθηκαν οι αλγόριθμοι σε πρότυπο σύστημα κ.λ.π.
5. ...

1.2 Structure of the Thesis

Εδώ περιγράφουμε τα κεφάλαια της διπλωματικής: μία πρόταση για το τί θα έχει κάθε κεφάλαιο. Συνήθως η ενότητα αυτή έχει την παρακάτω μορφή (δεν θα σας πάρει πάνω από μία μεγάλη παράγραφο):

Εργασίες σχετικές με το αντικείμενο της διπλωματικής παρουσιάζονται στο Κεφάλαιο 2. Το Κεφάλαιο ... συζητά θέματα μοντελοποίησης. Στο Κεφάλαιο ... αναπτύσσουμε κ.λ.π.

Για την τελική οργάνωση του κειμένου σας, συμβουλευθείτε τον επιβλέποντα της εργασίας.

Chapter 2

Background

2.1 Introduction

2.1.1 The Scope of Predictive Maintenance

There is a tremendous abundance of applications where the vibration analysis is a prerequisite. Readings and records are of vital importance in the context of the predictive maintenance implementation. Most of the industries nowadays strive to make use of accurate preventive maintenance plans, which requires a well-rounded continuous monitoring system, that keeps data of the whole equipment. The ultimate goal of the preventive maintenance plan is the elimination of the unwanted shut-downs due to a machine failure and the increase of plant reliability.

Most of the machines need bearings because bearings are mechanical components designed to support rotating shafts and minimize friction. The bearings failure may have a wide variety of causes, lubrication shortage, seal failure, high temperatures, misalignment are some of the causes among others. Therefore vibration analysis focuses on predicting all these possible causes of bearings failure. Some of the preventive actions against the bearings failure are described below.

Proper lubrication plays a vital role in bearing longevity. Rather than choosing oils based solely on moderate operating temperatures, engineers must consider factors like bearing speed, size, and load. Maintaining an adequate oil film is critical, and minimum viscosities are established based on bearing type. This oil film ensure that no metal-to-metal contact takes place for a predetermined load and speed. If these thresholds aren't met, metal-to-metal

contact occurs, significantly reducing reliability [1].

Alignment is another critical cause of bearings failure. A misalignment of some equipment -typically consisted of a motor and the main equipment- leads to bearings failure. For this purpose vibration and temperature monitoring are significant. For this reason, alignment practice upon machine installation is mandatory. According to Bloch the proper initial shaft alignment is necessary in any case and it is proven that the most advantageous technique to do so is the laser optic one [1].

A detailed alignment procedure is also presented by Bloch. On Figure 2.1 is depicted a typical aligning tool set-up. The laser headers must be installed on machined surfaces, which entails that it will be settled on the motor shaft and the shaft of the rotated mechanism. This laser mechanism has an emitter system, that produces a laser beam, and the laser beams are received by some receivers. Usually a three reference point alignment is what the optic laser machines offer, and this procedure takes place while the motor and the rotated machine are coupled with each other.

The laser mechanism set-up (laser emitters and receivers) are connected to some alignment computer, which is powered by some special software. This yields some reading and some alignment correction recommendations. The alignment criteria depend on the mechanical design of the equipment and the rotating speed as well [1].

However there could be some more causes of machinery failure, specifically a loose machinery base or tightening bolt is another that could lead to a components failure. Hence, the way to alleviate this issue is to align and tighten the machinery back to the frame.

How does vibration diagnosis contribute to operational reliability and cost efficiency in engineering applications? The answer to this question is that the implementation of vibration-analysis-based predictive maintenance helps in the direction of troubleshooting the cause of some issue. In addition, raising awareness on some upcoming failure leads to proper planning, which can prevent greater equipment damage.

2.1.2 The Lack of Preventive Maintenance

It is worth mentioning that there is a large number of possible faults that may lead to vibration, when the preventive maintenance is not implemented properly. The most common defect that is encountered on machinery is this of the bearing failure (see Figure 2.2 some bearings components and on Figure 2.3 a bearing's component defect).

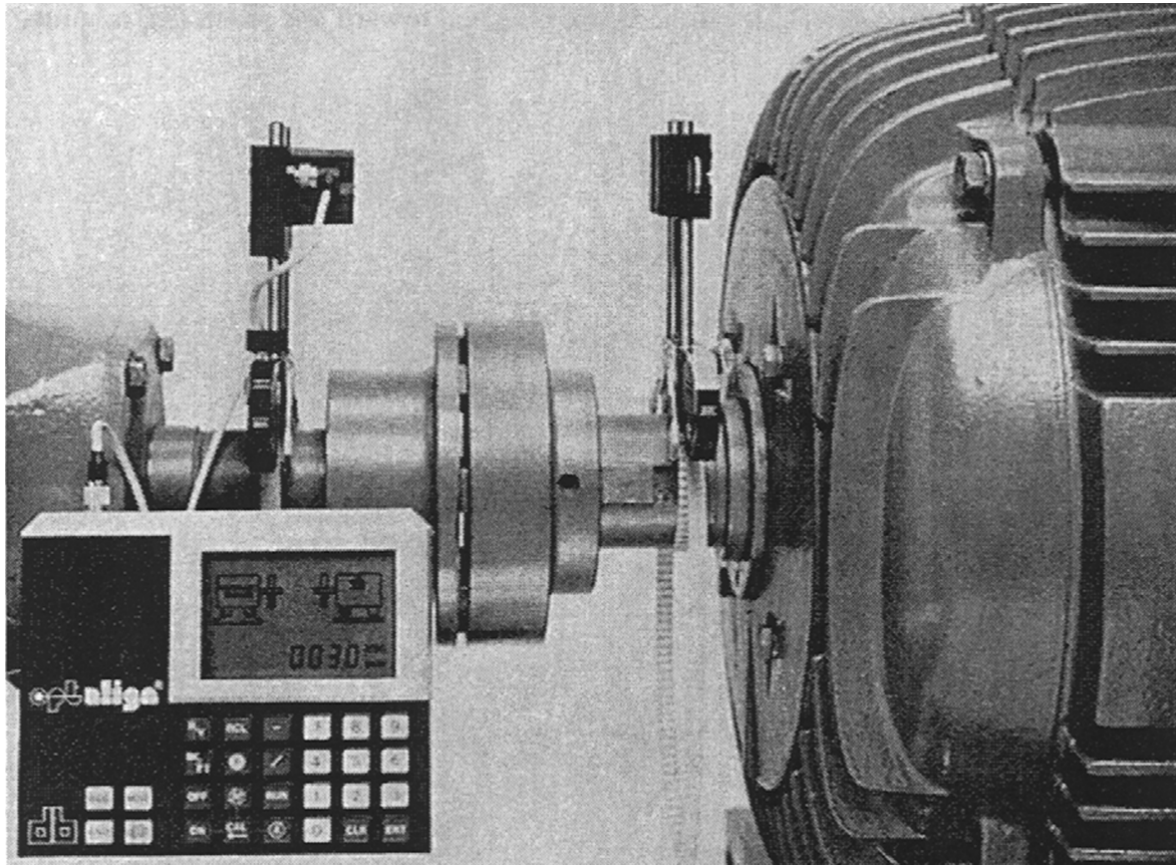


Figure 2.1: The Internet of Things paradigm as a result of the convergence of different visions. Source: [1].

A bearing's defective component will certainly lead to greater problems. Specifically, this could lead to a severe shaft wearing or even thread damage (see 2.4). In some other cases this could lead to a bearing housing damage.

Chen et al included in their research some more photos related to faulty bearings and gearsets [3].

2.1.3 The Reliability

To begin with, the ultimate objective of the preventive maintenance concept is to improve the dependability. In essence, it is a concept that encompasses different subconcepts like Reliability, Availability, Maintainability and others. The reliability interprets the successful uninterrupted operation of an adequately-performing device or an engineering system. This entails that the operating system must retain not only a low Failure rate, but the most important trait of a device in operation must be the absence of failures so as to claim the title of a "reliable system".

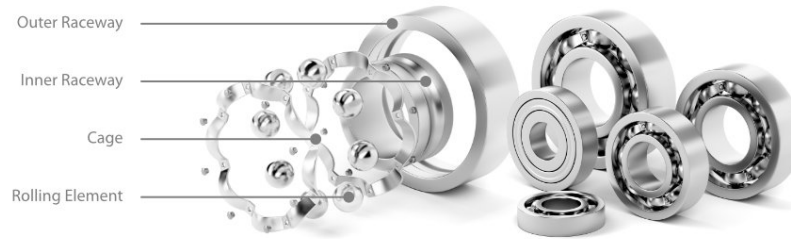


Figure 2.2: Different bearings components (source: [2]).



Figure 2.3: A bearing's defective outer raceway.

According to Martz, "reliability analysis is the engineering and statistical methodology used to assess the unknown reliability of a device. It may be used qualitatively as a means for identifying the various failure modes and causes, or it may be applied quantitatively, using appropriate failure data, to provide a numerical estimate of the unknown reliability" [9]. Hence, it is worth it utilizing the aforementioned metrics (MTTF and MTBF), when contemplating reliability, since these constitute the actual metrics of reliability.

The initial aspect of the theoretical background to be elaborated upon is Mean Time to Failure (MTTF) is a fundamental reliability metric used to estimate the average operational lifetime of non-repairable components and systems. In the context of this very article, the main focus of the study is mechanical components (e.g bearings, gears) as well as a system-machine in operation.

Some of the most important functions that are being used in the context of the metrics interpretation are the following:

- $f(x)$ is the probability density function of the time to failure.



Figure 2.4: Possible defects of shafts.



Figure 2.5: Faulty bearings and gears (source: [3]).

- $R(t)$ represents the probability that the device is still operational at time t [9].
- $F(t)$ is the cumulative distribution function of the time to failure, representing the probability that the system has failed by time t .

Interpretation of the MTTF

In this regard the MTTF can be modeled as the integral of the reliability function $R(t)$ over time, from zero to infinity:

$$\text{MTTF} = \int_0^{\infty} R(t) dt$$

Where:

$$R(t) = \Pr(T > t) = \int_t^{\infty} f(x) dx = 1 - F(t)$$

The MTTF interprets the expected time until failure for a large number of identical devices under similar operating conditions.

MTTF is used to predict the average lifespan of a device, providing critical information for maintenance planning, warranty determination, and lifecycle management. A higher MTTF indicates a more reliable device with a longer expected operational life. This metric is crucial for designing reliable products and for making informed decisions about product development and quality assurance.

Interpretation of the MTBF

The MTBF is another critical metric used in reliability engineering, particularly for repairable systems. It represents the average time between failures for a system or component. For repairable systems, MTBF is calculated as the sum of the operational time divided by the number of failures:

$$\text{MTBF} = \frac{\text{Total operational time}}{\text{Number of failures}}$$

In this case, the reliability function would be:

$$R(t) = e^{-\lambda t}$$

Mathematically, MTBF is the inverse of the failure rate λ for systems with a constant failure rate [9]:

$$\text{MTBF} = \int_0^{\infty} t f(t) dt = \int_0^{\infty} R(t) dt = \frac{1}{\lambda}$$

MTBF helps in understanding the reliability of a system over time and is essential for planning maintenance schedules, improving system design, and ensuring continuous operation with minimal downtime. A higher MTBF indicates a more reliable system that experiences fewer failures over its operational life.

Interpretation of the Failure Rate

The Failure rate, often denoted as λ , is defined as the frequency with which an engineered system or component fails. This metric is typically expressed in terms of the number of failures per unit time.

In mathematical term the Failure rate can be expressed as follows:

$$\lambda(t) = \frac{f(t)}{R(t)} = \frac{f(t)}{1 - F(t)}$$

There is three failure rate cases that may be encountered:

- **Constant Failure Rate:** A constant λ suggests that the system is in its "useful life" period, where the rate of failure is steady and does not change over time.
- **Increasing Failure Rate (IFR):** An increasing $\lambda(t)$ indicates that the system is aging and becoming more prone to failure.
- **Decreasing Failure Rate (DFR):** A decreasing $\lambda(t)$ is often observed during the initial "infant mortality" period, where early failures occur, but the remaining items are more reliable over time.

Constant Failure Rate

For systems with an exponential distribution of failure times (i.e., a constant failure rate), the failure rate λ is constant over time:

$$\lambda = \text{constant}$$

In this case, the reliability function $R(t)$ and the probability density function $f(t)$ are given by:

$$R(t) = e^{-\lambda t}$$

$$f(t) = \lambda e^{-\lambda t}$$

Interpretation of the Availability

Availability (A) is typically expressed as a percentage and is calculated using the following formula[10]:

$$A = \frac{MTBF}{MTBF + MTTR} \times 100\%$$

Where:

- MTBF (Mean Time Between Failures): The average time interval between two consecutive expected failures of a system component or mechanism.
- MTTF (Mean Time To Repair): The average time required to repair a failed system component or mechanism and restore it to operational status.

2.1.4 Vibration Analysis Instruments

There is a great number of vibration monitoring products. Some of them are listed here-under:

- amplitude velocity and acceleration (range of motion) vibrometers
- frequency (rate of oscillation repetition) vibrometers
- accelerometers

The accelerometers and vibrometers, both can be remote - cordless sensor units or wired sensor units connected to some computer. The latter device is a smart device powered by some software so as to store and analyze data. In many cases these devices may also offer some online-cloud options, storing data on some database yielding accessibility for multiple users.

The device depicted in the image exemplifies a standard configuration of a vibration meter and an accelerometer (see Figure 2.6).

The accelerometer measures vibration events in mg (or in some cases in mm/s^2 and interprets these events into frequency domain data. While the vibrometer stores reading of vibration velocity (mm/s) of vibration acceleration (mm/s^2).

These devices have the following common traits, they require the machinery size and rotation as an input, so as to proceed with the results extraction. In addition to this, these can execute this calculation on all three directions simultaneously. And to top it all off, they are expensive.

A deeper understanding of the available in the market sensor is given by Lilly et al [11]. They divide vibration equipment into three levels of detail:

- hand-held sensor
- single-channel FFT

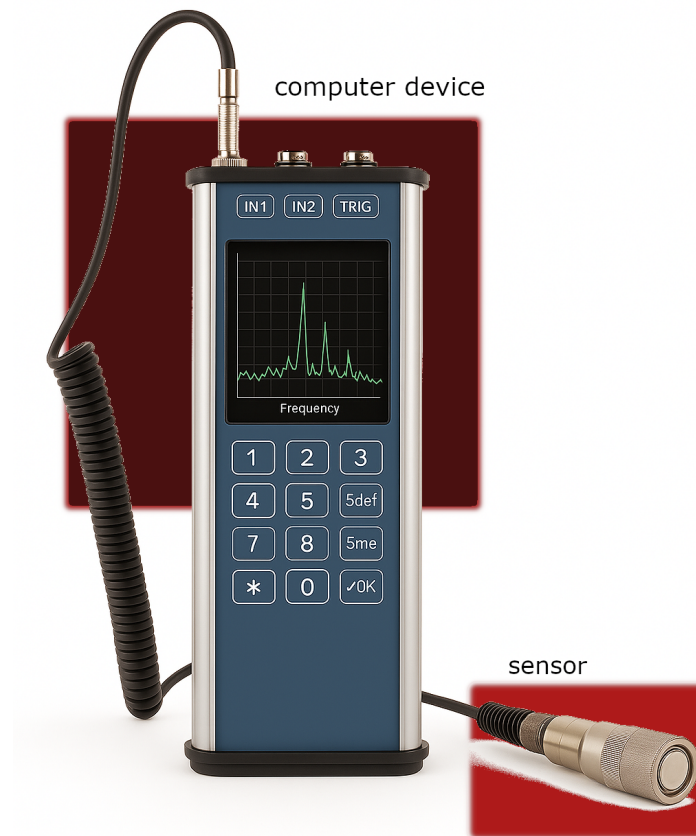


Figure 2.6: Typical vibrometer set-up.

- multi-channel FFT

Vibration measurement tools vary significantly in their technical capabilities. Basic hand-held devices offer convenient portability for field use but have several limitations. These compact units are confined to single-location measurements and typically operate within a 10-1000 Hz frequency window. Using velocity transducers, they provide only fundamental vibration amplitude data without phase information or sophisticated analytical features.

More advanced single-channel FFT instruments deliver superior performance for spectral examination. These systems cover a broader 0.5-20 kHz frequency span with enhanced resolution, supporting up to 1600 spectral lines. While they enable more thorough vibration spectrum evaluation, their single-input design prevents multi-point correlation studies or phase-based diagnostics.

For comprehensive vibration analysis, multi-channel FFT platforms are necessary. These high-performance systems accommodate multiple measurement points simultaneously across an extensive DC-100 kHz bandwidth. With ultra-fine resolution reaching 6400 spectral lines, they capture precise phase relationships vital for resonance detection and operational deflection pattern visualization. Such systems incorporate specialized functions including rotational speed tracking, coherence verification, and system response characterization - all crucial for detailed mechanical fault investigation and structural dynamic studies.

The primary technical distinctions between these solutions involve: their measurement approach (single versus multiple points), frequency analysis quality (basic versus high-definition), phase measurement availability, and specialized analytical features. Choosing the appropriate tool involves matching the instrument's capabilities to the diagnostic requirements - from preliminary condition checks with hand-held units to in-depth failure analysis using multi-channel systems.

There is also some obsolete set-ups, for instance the station used by Betta et al for their experiment [4]. This station encompasses an acceleration sensor (8710A50M1 by Kistler™) with 5g measuring range, a signal conditioner with adjustable gain (5118B1 by Kistler™) for data translation, a DPC/C40 by Loughborough Sound Images™ for FFT-Analysis and a host PC (see Figure 2.7).

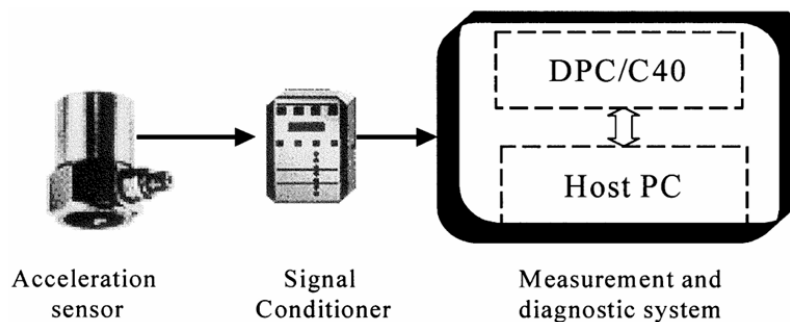


Figure 2.7: Block diagram of the designed station for vibration analysis (source: [4]).

There is some more accessible alternatives. Specifically, there is some small sized low-cost acceleration sensor on breakouts that can be used on related projects (see Figure 2.8).

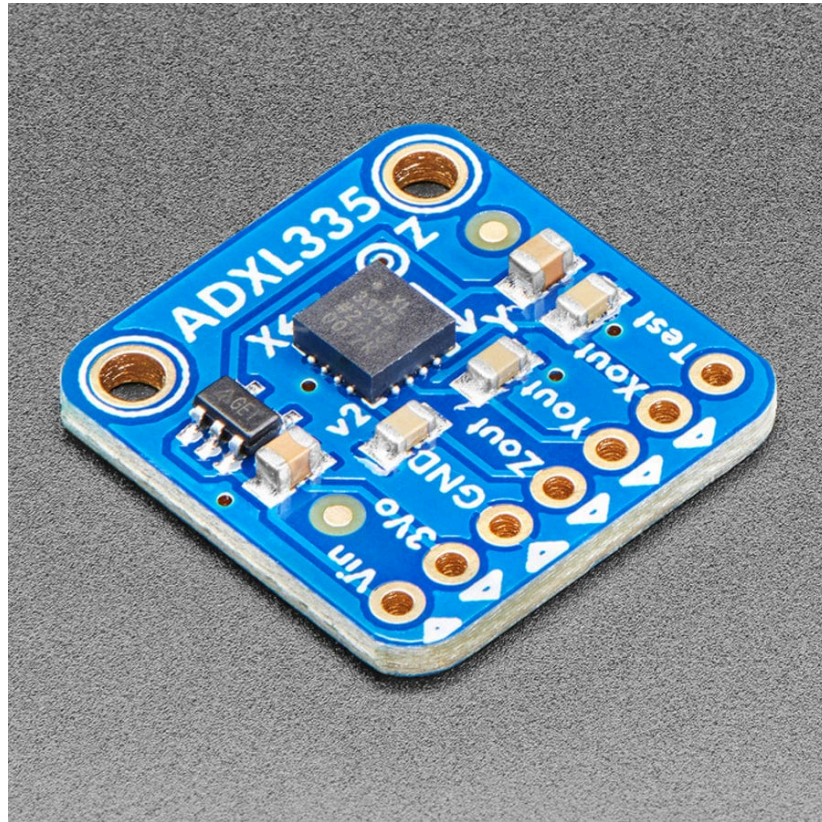


Figure 2.8: ADXL 335 accelerometer.

2.1.5 Typical Applications of Vibration Analysis

Some typical applications that use vibration analysis implementations are turbine motors and shafts, motors connected pumps and motors connected to roll equipment among others.

Li and McKee explain in their research that most of the vibration analysis applications are related to electrical and mechanical equipment that utilize roller bearings [12].

Similarly, Lilly and Marscher have studied the vibration and noise derived from pumps and motors separately [11]. In addition, the concept of translational and torsional vibration is introduced. In addition to this, the concept of vibration analysis on piping is studied as well. In this context, the translational vibration analysis is made for the engine-pump-support structure, investigating the lateral vibration. For this study the usage of high-quality sensors is a prerequisite. There is also the concept of the torsional vibration analysis, which is also subtle and requires special study on the oscillating torque.

In addition to the aforementioned study, Lilly and Marscher investigate also the possibility of using some actuators that may compensate the noise of the motor-pump compound, making the most of the electronically generated sound waves. This technique aims at the cancellation

of noise pollution [11].

The study by Jiao et al. demonstrates the efficacy of time-domain feature extraction combined with Independent Component Analysis (ICA) for gearbox fault diagnosis. Their approach maximizes mutual information to isolate fault-related vibrations from noisy multi-channel measurements, achieving high accuracy in classifying faults such as teeth surface damage and loose foundations [13].

There is also the chance of implementing advanced signal processing techniques, such as envelope analysis and cyclostationary methods, to enhance fault detection in complex systems like helicopters, where high noise and low signal-to-noise ratios are prevalent [14].

In the context of unmanned aerial vehicles (UAVs), Mohd Ghazali and Rahiman employ vibration sensors and artificial intelligence (AI) techniques, including fuzzy logic and neural networks, to detect faults in drone arms. Their framework addresses the challenges of real-time decision-making and integrates IoT for remote monitoring, showcasing the adaptability of vibration analysis to emerging technologies [15]. Meanwhile, Girondin et al. focus on bearing faults in helicopters, proposing redundancy-based harmonic analysis to overcome limitations imposed by low sampling frequencies and environmental noise. Their geometric mean criterion proves robust in identifying fault frequencies despite parasitic vibrations [14].

Theoretical advancements in vibration analysis also emphasize the role of preprocessing steps, such as whitening and Minimum Entropy Deconvolution (MED), to accentuate impulsive features indicative of faults [14]. These methods, combined with machine learning classifiers like Support Vector Machines (SVM) and neural networks, enable automated fault diagnosis with minimal human intervention, as demonstrated in gearbox and bearing applications [13, 15].

Vibration analysis serves as a versatile tool across industries, from traditional rotating machinery to cutting-edge UAVs. Its evolution—driven by innovations in signal processing, AI, and IoT—continues to expand its applicability and precision in fault detection and predictive maintenance.

2.2 Fundamentals of Bearing Fault Frequencies

Li and McKee analyze the failure modes of bearing and yield a brief explanation of what each failure mode entails. It is highlighted by their research that wide-band impulses are

generated when rollers pass over a defect. As far as the diagnostics algorithm is concerned, in this research there is presented statistical parameters that help in finding bearings localized defects. In addition to this, they stress that the frequency-related algorithms like the Hilbert transform are the most prominent in bearing diagnostics.

The most significant challenge that Betta et al had to deal with is that of the faulty cases detection. For this reason, fault and unfault models have been developed, and series of comparisons between the actual model generated by apparatus measurements and these two situational models are being executed. This development constitutes a real challenge since this is the main criterion that identifies the state of the motor [4].

2.2.1 Unfault and Fault Models

Betta et al make use of a model-based approach, which entails that the main concept is comparison-oriented. Specifically, there is the unfault model, which contains a high-amplitude peak at the shaft rotating frequency F_ω . In addition to that, it can be noticed in Figure 2.9 that some other patterns distinct themselves from the previously-mentioned unfault model. Regarding the unbalance model, it is obvious that the amplitude at the first tone is much greater. In the case of the misalignment model, it can be noticed that the tone at $2F_\omega$ has a much greater amplitude while the amplitude at the tone of the shaft speed F_ω remains at a high level. In the case of looseness, there are some more tones prevailing on the spectrum, among the tones related to the shaft speed. To top it all off, the most important vibration spectrum is that of the bearing failure (Refer to Figure 2.10). Every bearing is characterized by four critical frequencies, given by its manufacturer, and these frequencies are associated with the four bearing components (inner and outer ring, balls, and cage) [4]:

- BPFO (Ball Pass Frequency Outer) - outer race failure
- BPFI (Ball Pass Frequency Inner) - inner race failure
- BSF (Ball Spin Frequency) - rolling element failure
- FTF (Fundamental Train Frequency) - cage failure

The procedure to calculate critical frequencies involves two steps: first computing the bearing frequency factors, then deriving the vibration frequencies. The following analysis covers the case where the inner race rotates while the outer race remains stationary [12], [4].

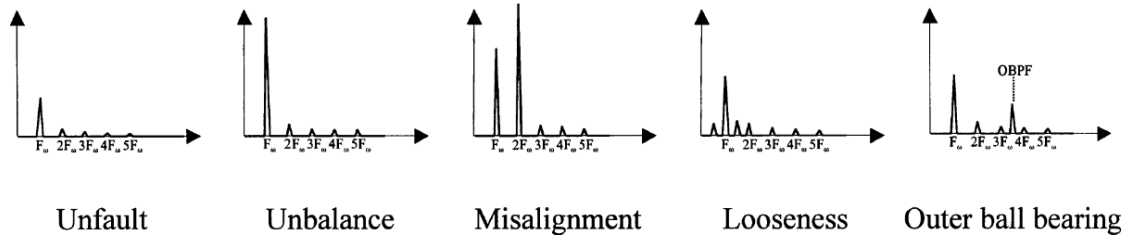


Figure 2.9: Fault and unfault spectrum graphs (source: [4]).

$$F_{OR} = \frac{Z}{2} \left(1 - \frac{d}{D} \cos \alpha \right) \quad (\text{Outer Race Fault}) \quad (2.1)$$

$$F_{IR} = \frac{Z}{2} \left(1 + \frac{d}{D} \cos \alpha \right) \quad (\text{Inner Race Fault}) \quad (2.2)$$

$$F_{FTF} = \frac{1}{2} \left(1 - \frac{d}{D} \cos \alpha \right) \quad (\text{Cage Frequency, Inner Race Rotating}) \quad (2.3)$$

$$F_{BS} = \frac{D}{d} \left[1 - \left(\frac{d}{D} \right)^2 \cos^2 \alpha \right] \quad (\text{Ball Spin Frequency}) \quad (2.4)$$

Then, the aforementioned factors when multiplied with shaft speed (f) give specific critical bearing vibration frequencies:

$$BPFO = f \times F_{OR} \quad (2.5)$$

$$BPFI = f \times F_{IR} \quad (2.6)$$

$$BSF = f \times F_{BS} \quad (2.7)$$

$$FTF = f \times F_{FTF} \quad (2.8)$$

where:

f	: Shaft rotational speed (Hz)
BPMI	: Ball pass frequency, inner race
BPMO	: Ball pass frequency, outer race
FTF	: Fundamental train frequency
BSF	: Ball spin frequency
Z	: Number of rolling elements
D	: Pitch circle diameter of the bearing (mm)
d	: Rolling element (ball) diameter (mm)
α	: Contact angle ($^{\circ}$)

It is also worth noting that the bearing frequency factors can be obtained through:

- Manufacturer-provided values
- Direct calculation using the above equations (2.1)–(2.4)
- Reverse calculation from measured vibration frequencies (2.5)–(2.8)

To summarize the preceding analysis, Figure 2.10 provides a clear visualization of the fault diagnosis procedure. When analysis reveals elevated frequency tones, a systematic approach must be followed to identify the source of these dominant frequencies. This process essentially involves mapping characteristic vibration patterns to specific machine faults on a frequency spectrum plot.

It must be that a fault-emulation method can be used in such cases to simulate the faulty models. This entails that additional tones can be stressed, and crucial spectra modifications may take place for research and comparison purposes.

2.3 Algorithms and Technologies

2.3.1 Algorithm Introduction

This section aims at yielding a brief introduction into the concept of bearing failure predictive maintenance plan. The set-up utilizes an accelerometer which collects data that are

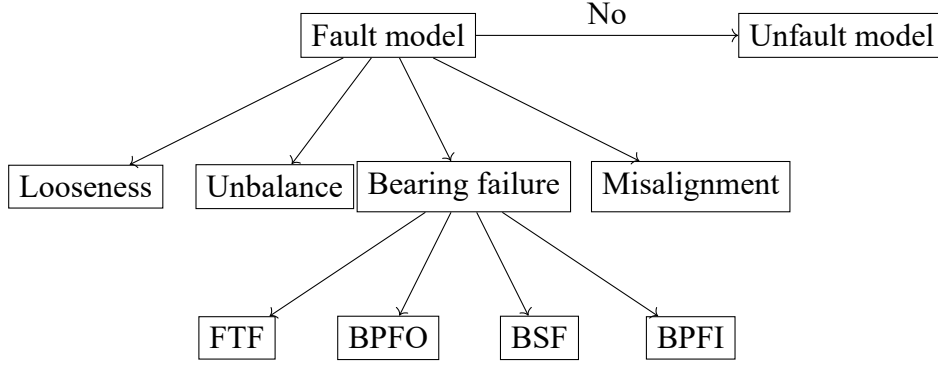


Figure 2.10: Fault tree analysis diagram showing the relationship between machine conditions and specific fault frequencies.

going to be used for model assessment, by the means of Fast-Fourier Analysis.

In most of the actual applications a machine learning implementation is a common practice. A figure that depicts this modeling has been presented by Dharmarathne et al [5].

2.3.2 Signal Processing

Fourier Analysis

In this section a deeper elaboration on Fourier Analysis and Transform is going to take place.

Serafeimidou explains in detail the theoretical background of the Fourier Analysis [16]. For an arbitrary finite range $[-L, L]$ there is a family of functions, which meet the Dirichlet condition, and these functions can be expressed as a trigonometric sequence:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\pi\omega x) + b_n \sin(n\pi\omega x)), \quad (2.9)$$

where $\omega = \frac{\pi}{L}$.

In addition its constants can be expressed as follows:

$$a_n = \frac{1}{L} \int_{-L}^L f(u) \cos(n\omega u) du \quad (2.10)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(u) \sin(n\omega u) du \quad (2.11)$$

The equation (2.9) needs to be expressed as a valid form within the whole $x \in \mathbb{R}$. For this reason the trigonometric integral (which is called Fourier integral) can be expressed as

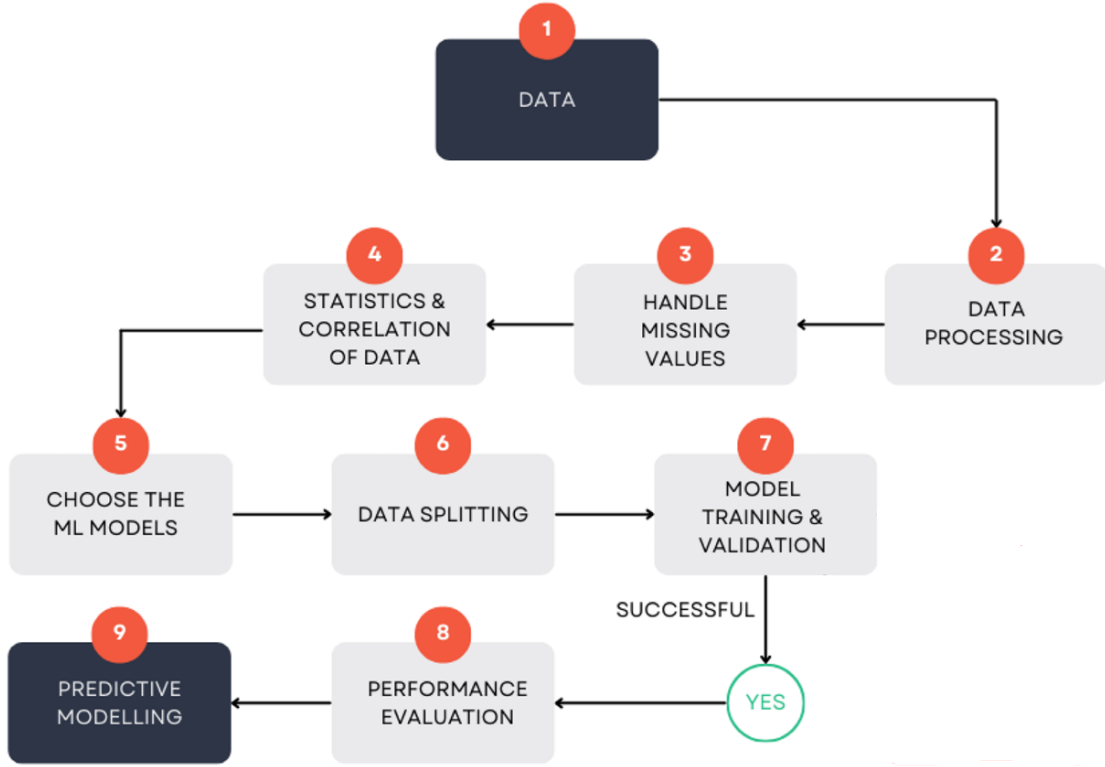


Figure 2.11: Data collection and assessment algorithm for bearing failure prediction. Source: [5].

follows:

$$F(x) = \int_0^{\infty} [A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)] d\omega \quad (2.12)$$

where:

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \cos(\omega u) du \quad (2.13)$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \sin(\omega u) du \quad (2.14)$$

Assuming that there exist real functions $f(x)$ defined in $\mathbb{R}$ meeting the following points:

- $f(x)$ and its derivative $f'(x)$ are continuous in the range $[-L, L] \subset \mathbb{R}$.
- $f(x)$ is integrable in $\mathbb{R}$:

$$\int_{-\infty}^{\infty} |f(x)| dx < \infty.$$

Hence, given that the aforementioned conditions are met, $A(\omega)$ (see Eq. (2.13)) and $B(\omega)$ (see Eq. (2.14)) tend to attain a real value for ω , which implies that $A(\omega)$ and $B(\omega)$ are defined

for every $\omega \in \mathbb{R}$. In addition to this, the integral of the equation (2.12) tends to attain a real value for x , which ensures that $F(x)$ is defined for every $x \in \mathbb{R}$.

$F(x)$ is identical to $f(x)$ whenever $f(x)$ is a continuous function. There are some special cases when $f(x)$ is not a continuous function, and to deal with these cases, we define:

$$F(x) = \frac{1}{2} \left[\lim_{x \rightarrow x^+} f(x) + \lim_{x \rightarrow x^-} f(x) \right] \quad (2.15)$$

This ensures that $F(x)$ is properly handled even in cases of discontinuity.

All things considered the Fourier Transform $F(\omega) = \mathcal{F}\{F(x)\}$ can be mathematically expressed as follows:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (2.16)$$

where the frequency can derive from the following form: $\omega = \frac{2\pi jk}{N}$.

In practice, the Discrete Fourier Transform (DFT) boasts an extremely wide variety of applications, facilitating the Fourier Transform when applied to sampled signals.

Hence, the mathematical formulation for determining the amplitude of each frequency, given a waveform $x_0, x_1, \dots, x_{n-1}$ consisting of real values, is structured as follows:

$$X(j) = \sum_{k=0}^{n-1} x(k) e^{-\frac{i2\pi jk}{n}}, \quad \text{for } j = 0, 1, \dots, n-1. \quad (2.17)$$

Fast Fourier Transform (FFT)

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the Discrete Fourier Transform (DFT), expressed by the equation (2.17) and its inverse. One of the most prominent FFT algorithms, the Cooley-Tukey algorithm [17], employs a divide-and-conquer approach to reduce the computational complexity from $\mathcal{O}(n^2)$ (for the naive DFT) to $\mathcal{O}(2n \log n)$, enabling significant speedups for large amount of data.

This reduces complexity from $\mathcal{O}(n^2)$ (naive DFT entails direct implementation of the equation (2.17)) to less than $\mathcal{O}(2n \log n)$ (a Fast Fourier Transform implementation), enabling major time savings for complex and large-scale transforms. It is also worth articulating that the complexity can be interpreted as the number of operations executed [17], [18].

Both algorithms are implemented in major numerical packages including MATLAB and NumPy's routines [19]. What is highlighted by Singleton is that the Cooley-Tukey approach

is considerably faster and decently performing especially when coupled with some additional practices so as to execute the Fast Fourier Transform [18].

The computation time according to the aforementioned article decreases drastically by making use of the Cooley-Tukey algorithm. In addition, this very algorithm performs and adapts better to different applications compared to Good's, Danielson's and Goertzel's approaches.

Cooley and Tukey proved that the sample length can be composite size where it is preferable to make use of sample points sized as power of 2, especially when it comes to personal computers with binary arithmetic system. In addition to this the authors have proven that the power of 3 is formally the most efficient approach, however in the context of the fast Fourier Analysis the so-called Radix-2 implementation is the prominent one [17].

$$N = 2^m \quad (2.18)$$

It is hereby worth it elaboration on the Fast Fourier Transform algorithm [17]. The indices of the equation (2.17) can be expressed as follows:

$$\begin{aligned} j &= j_{m-1} \cdot 2^{m-1} + j_{m-2} \cdot 2^{m-2} + \dots + j_1 \cdot 2 + j_0, \\ k &= k_{m-1} \cdot 2^{m-1} + \dots + k_1 \cdot 2 + k_0, \end{aligned} \quad (2.19)$$

where j_v and k_v are the contents of the respective bit positions within the binary representation, which ensues that they are equal to 0 or 1.

The aforementioned set-up (see eq. (2.17)) leads to the following equation (2.21), encompassing twiddle factors expressed as:

$$W = e^{-2\pi i/N} \quad (2.20)$$

And the Fast Fourier Transform:

$$X(j_{m-1}, \dots, j_0) = \sum_{k_0} \sum_{k_1} \dots \sum_{k_{m-1}} x(k_{m-1}, \dots, k_0) \cdot W^{j \cdot k_{m-1} \cdot 2^{m-1} + \dots + j \cdot k_0} \quad (2.21)$$

where as per the first step:

$$W^{j \cdot k_{m-1} \cdot 2^{m-1}} = W^{j_0 \cdot k_{m-1} \cdot 2^{m-1}} \quad (2.22)$$

Furthermore, in the context of the first calculation, the first array can be expressed as follows:

$$x_1(j_0, k_{m-2}, \dots, k_0) = \sum_{\mathbf{k}_{m-1}} x(\mathbf{k}_{m-1}, \dots, \mathbf{k}_0) \cdot W^{j_0 \cdot \mathbf{k}_{m-1} \cdot 2^{m-1}} \quad (2.23)$$

And hereunder can be contemplated that proceeding to the next innermost-successive sum:

$$W^{j \cdot k_{m-1} \cdot 2^{m-1}} = W^{(j_{l-1} \cdot 2^{l-1} + \dots + j_0) \cdot k_{m-l} \cdot 2^{m-l}} \quad (2.24)$$

followed by the calculation of the arrays:

$$x_l(j_0, \dots, j_{l-1}, k_{m-l-1}, \dots, k_0) = \sum_{\mathbf{k}_{m-1}} x_{l-1}(j_0, \dots, j_{l-2}, \dots, k_{m-l}, \dots, k_0) \cdot W^{(j_{l-1} \cdot 2^{l-1} + \dots + j_0) \cdot k_{m-l} \cdot 2^{m-l}} \quad (2.25)$$

for $l = 1, 2, \dots, m$

And this could constitute a recursive procedure obtaining the successive arrays. And last but not least, this finally could lead to the calculation of X spectrum. It can be contemplated that from the Equation (2.25), there is at all times two consecutive values, the even and the odd value that can be stored at a specific location corresponding to an index of:

$$(j_0 2^{m-1} + \dots + j_{l-1} 2^{m-l} + k_{m-l-1} 2^{m-l-1} + \dots + k_0) \quad (2.26)$$

It is also worth it recalling that m is defined by $N = 2^m$. Since bit positions $2^{(m-l)}$, meaning the j and k coefficients, can only get the values 0 and 1, which ensues that there is multiple-storage locations that can be calculated simultaneously.

The array calculated gives the desired Fourier sums:

$$X(j_{m-1}, \dots, j_0) = A_m((j_0, \dots, j_{m-1})) \quad (2.27)$$

At this point there is a need to simplify the Fast Fourier Transform algorithm. It is mandatory to recall the (2.21) so as to depict the algorithm.

The calculations workload can be reduced, by decomposing the original input sequence $X[k]$, of length N , into two subsequences:

- A sequence consisting of elements at even-numbered indices: $X_{\text{even}}[k]$
- A sequence consisting of elements at odd-numbered indices: $X_{\text{odd}}[k]$

By processing these smaller sequences independently, the algorithm substantially reduces the computational workload. The FFT is applied recursively to each half, and this decomposition continues until each subproblem has length 1, at which point the computation becomes trivial (the amplitude value corresponding to a specific frequency). Furthermore, the results from the smaller FFTs are recombined using twiddle factors (see Equation (2.20)).

An significant property of this algorithm is that that, in the case of $j > \frac{n}{2}$, the values are repeated due to the sinusoidal property of $\exp$, which is known as the symmetry identity. This entails that for the values $j = 0, 1, \dots, n/2 - 1$:

$$X_{\text{even}}(j + n/2) = X_{\text{even}}(j) \quad \text{and} \quad X_{\text{odd}}(j + n/2) = X_{\text{odd}}(j) \quad (2.28)$$

The evens and odds decomposition:

$$X_{\text{even}}(j) = \sum_{k=0}^{n/2-1} x(2k) e^{-\frac{i2\pi jk}{n/2}}, \quad \text{for } j = 0, 1, \dots, n/2 - 1. \quad (2.29)$$

$$X_{\text{odd}}(j) = \sum_{k=0}^{n/2-1} x(2k+1) e^{-\frac{i2\pi jk}{n/2}}, \quad \text{for } j = 0, 1, \dots, n/2 - 1. \quad (2.30)$$

So from the Equation (2.21):

$$X(j) = X_{\text{even}}(j) + e^{-\frac{i2\pi j}{n}} X_{\text{odd}}(j) \quad (2.31)$$

And this procedure is scalable since each and every X_{even} and X_{odd} can be depleted into a subsequent set of X_{eve} and X_{odd} to the point that the length 1 is reached.

Last but not least, the symmetry identity as described by the Equation (2.28), leads to the following set of equations (see also (2.31)):

$$X(j + n/2) = X_{\text{even}}(j) + e^{-\frac{i2\pi(j+n/2)}{n}} X_{\text{odd}}(j) \quad (2.32)$$

For the values $j = 0, 1, \dots, n/2 - 1$.

The objective of this algorithm is to get to the point that the length is of the sums is 1, meaning that the sums of the Equations (2.29) and (2.30) are applicable for $k=0$. The only remaining exponential terms $e^{-\frac{i2\pi j}{n}}$ in the Equation (2.31) - as previously described by Equation (2.25).

The aforementioned procedure can be further repeated, by breaking the branches problem into sub-branches. The radix-2 DIT algorithm to be formulated as follows on Figure 2.12, depicting the concept of narrowing down the problem.

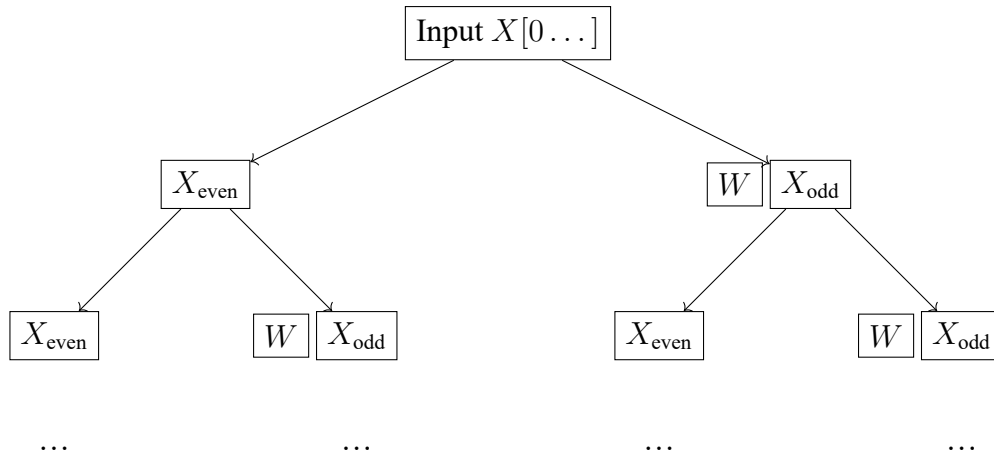


Figure 2.12: Recursive decomposition using even-odd splitting.

All in all, this section yields an overview of utmost importance highlighting the most invaluable Fourier Transform algorithms expressed by equation (2.17) and equation (2.21). As discussed in this section, the Cooley-Tukey Fast Fourier Transform (FFT) algorithm significantly accelerates the computation of the Discrete Fourier Transform (DFT) by recursively breaking the problem into smaller, more manageable subproblems. This divide-and-conquer approach is fundamental to the algorithm's performance advantage.

2.4 Internet of Things

2.4.1 Introduction to Internet of Things & Key Technologies

The ultimate objective of this project is to create a set-up that is well-rounded and has a real-time monitoring and assessment competency. Two of the most prominent technologies that are being incorporated by Internet of Things (IoT) is the cloud computing and the wireless communication. It is also worth mentioning that, Internet of things encompasses interconnection of physical devices and applicaitons making the most of sensors, networks and data analysis.

2.4.2 Cloud-Oriented Perspective

An extremely accurate definition of the *Internet of Things* is given by Gubbi et al, articulating that:

“Interconnection of sensing and actuating devices providing the ability to share

information across platforms through a unified framework, developing a common operating picture for enabling innovative applications. This is achieved by seamless ubiquitous sensing, data analytics and information representation with Cloud computing as the unifying framework.”

This approach is mostly ”Cloud”-oriented. The authors focus mostly on the concept of ubiquitous information and evolving communication networks. The means to give ground to this very concept is Smart Connectivity. It is fueled by the prevalence of devices enabled by open wireless technology such as Bluetooth, radio frequency identification (RFID), Wi-Fi, and telephonic data services as well as embedded sensor.

Nevertheless, Hardware-sensor breakouts are critically important, as they serve as the primary means of data collection in IoT systems. Middleware and computing tools for data analytics are equally invaluable, as they enable the processing of raw sensor data into actionable insights. Finally, visualization tools and dashboards play a key role in end-user interaction with the IoT environment, while also providing stakeholders with an intuitive and accessible representation of the data [20].

2.4.3 Internet- vs. Things-Oriented Perspectives

As regards the approach of Atzori et al, there is given a different definition by considering two different approaches, the ”Internet”-oriented approach and the ”Things”-oriented approach. The ”Things”-oriented approach targets mainly at Big Data applications, since according to Atzori et al the time, the place and multiple recipients are the most important factors of this approach. In essence the ”Things” that contribute to this perspective are the Radio-Frequency Identification (RFID), Near Field Communications (NFC) and Wireless Sensor and Actuator Networks (WSAN) among others.

The ”Internet”-oriented vision given by Atzori et al, it is highlighted that the IP (Internet Protocol) constitutes the core technology of Internet of Things (IoT). The Internet Protocol (IP) is lightweight, scalable, and capable of connecting smart objects worldwide—even on low-power devices. The authors also introduce Internet Ø, a similar approach that further simplifies IP to enable ”IP over anything,” ensuring seamless IoT integration. The ultimate goal of both strategies is to make all objects addressable, reachable, and interoperable through IP, moving from a basic ”Internet of Devices” to a fully connected ”Internet of Things” [6].

2.4.4 Semantic-Oriented Approach

In addition to the aforementioned perspectives there is an emerging need for representation and storage of the invaluable information, which leads to the "Semantic"-oriented approach (see 2.13). This final approach creates a well-rounded Internet of Things (IoT) concept, yielding more details on it.

The whole concept of the vast majority of different projects requires a semantic approach so as to set some thresholds and yield some feedback or take some action over something. In addition to this the semantic vision according to Atzori et al expresses the need for a representation of the collected data [6].

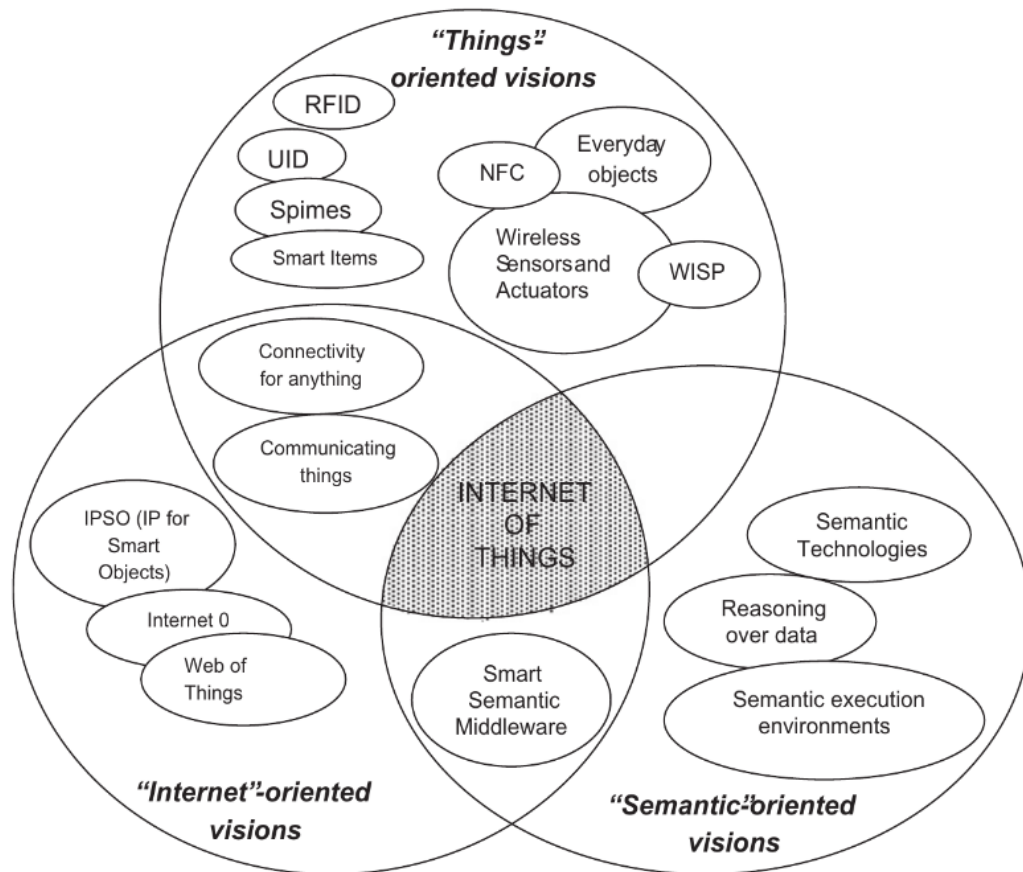


Figure 2.13: The Internet of Things paradigm as a result of the convergence of different visions. Source: [6].

All in all Atzori defines Internet of Things (IoT) as follows:

“Internet of Things constitutes a world-wide network of interconnected objects uniquely addressable, based on standard communication protocols.”

2.5 Project Implementation: Sensing & Communication

2.5.1 The Sensor Specifications

As it has been highlighted by the Internet of Things definitions, the sensors are among the most important devices of this technology. The sensing unit in this project utilizes an ADXL335 accelerometer to collect physical acceleration data, which connects to the micro-controller through an analog input interface.

The accelerometer ADXL335 is a single component sensing X, Y and Z, having a wide measuring range in all three axes. Typical applications of this sensor is structure deflection projects. This sensor is the ideal choice when it comes to vibration analysis applications, since it retains a measuring range of $\pm 3g$. This range adequate with respect to vibration analysis because in most of the rotating equipment applications an acceleration greater than 1,5g indicates a defective component. This entails that the $\pm 3g$ is sufficient in the context of this project. In addition to this, it offers higher resolution, up to 300mV/g for vibration detection, which means it has relatively high sensitivity to small vibrations. As regards its frequency bandwidth, it ranges 0.5-1600Hz. As regards its noise density, it equals to $150 \mu g/\sqrt{Hz}$, which is ideal for low-amplitude vibration waveforms[21].

The Adafruit ADXL335 is a high performance sensor breakout, utilizing capacitors so as to decouple the noise of the power supply and filters om analog outputs for antialiasing and noise reduction.

As for the sensor's theory of operation [21]:

ADXL335 is a polysilicon surface-micromachined structure built on top of a silicon wafer. Polysilicon springs suspend the structure over the surface of the wafer and provide a resistance against acceleration forces Deflection of the structure is measured using a differential capacitor that consists of independent fixed plates and plates attached to the moving mass. The fixed plates are driven by 180° out-of-phase square waves. Acceleration deflects the moving mass and unbalances the differential capacitor resulting in a sensor output whose amplitude is proportional to acceleration. Phase-sensitive demodulation techniques are then used to determine the magnitude and direction of the acceleration.

This combination of mechanical design and signal processing provides robust acceleration measurement without requiring temperature compensation circuits -since this breakout

retains a large tolerance into operation temperatures- or complex calibration procedures [21].

2.5.2 The Communication Protocol

The system architecture of the ADXL335 breakout supports analog communication protocol option, however there is a number of different communication protocols frequently utilized by the Adafruit breakouts:

- **Analog Input:** Currently implemented for direct sensor interfacing (ADXL335)
- **Digital Input:** Available for threshold-based detection applications or PWM
- **SPI (Serial Peripheral Interface):** Suitable for high-speed (up to 10 Mbps) communication with multiple peripherals
- **I²C (Inter-Integrated Circuit):** Enables multi-device communication using only two wires (SDA/SCL)
- **UART (Universal Asynchronous Receiver-Transmitter):** Implemented at a pre-determined baud rate for Arduino-to-Raspberry Pi communication

The Universal Asynchronous Receiver-Transmitter Protocol

The Arduino board is connected to the Raspberry Pi microcomputer via its USB port. The USB port of the Arduino is in essence a UART device, that has two options, not working simultaneously. The first one out of the two is the Rx - Tx pins option and the second one is the USB-cable option. In the case of the Arduino-Raspberry Pi interconnection, the desired set-up requires a USB-cable.

The Universal Asynchronous Receiver-Transmitter (UART) is a peripheral device for asynchronous serial communication. The baud rate and the data format is configurable, based on the project needs. After hands-on try and error, it has been found out that a baud rate of 115200 performs adequately offering a higher speed on data transferring, which is vital for the sake of this very project. The data transferring speed is of vital importance, since the higher the amount of data sent to the database is, the larger the frequency domain data range can be.

The raw UART (transmitted via USB) data is in a binary format, and upon receipt by the Raspberry Pi the data batch needs to be converted into a string format. For this reason it is

mandatory to use UTF-8 (8-bit Unicode Transformation Format) to handle ASCII characters at this point.

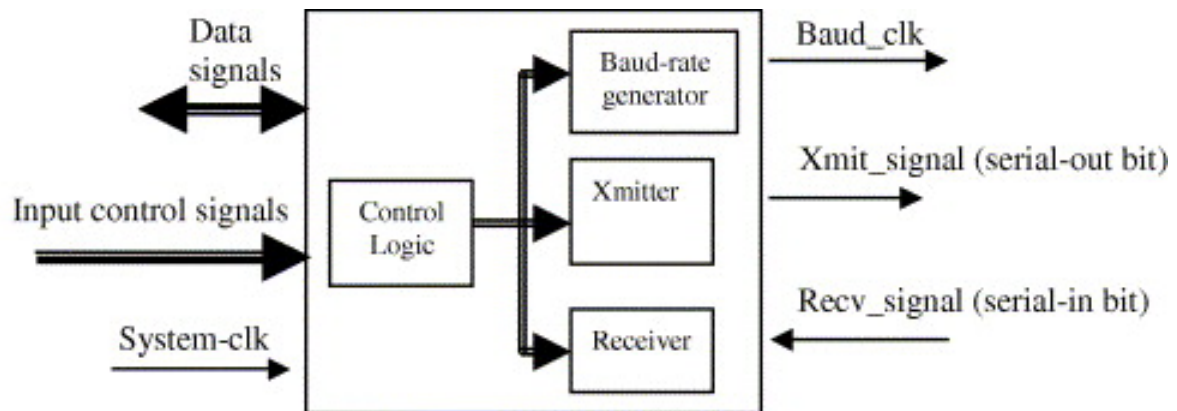


Figure 2.14: The UART block diagram. Source: [7].

A standard UART includes the following key modules (see Figure 2.14):

- **Transmitter** (parallel-to-serial conversion)
- **Receiver** (serial-to-parallel conversion)
- **Baud-rate generator**
- **Control logic**

The baud-rate generator derives a *baud clock* from the system clock. This clock typically operates at 16 times the desired baud rate:

$$\text{baud_clock} = 16 \times \text{baud_rate}$$

This oversampling approach ensures precise timing for bit detection and improves communication reliability.

UART transmits data in structured frames composed of the following elements:

- **Start bit:** Always logic 0
- **Data bits:** Typically 6 to 8 bits (configurable)
- **Optional parity bit:** For basic error detection
- **Stop bits:** Logic 1, usually 1 or 2 bits

The receiver “synchronizes” its internal clock to that of the transmitter’s at the beginning of every data packet received. This allows the receiver to sample the data bit at the bit-cell center. A data packet is composed of 1 start-bit, which is always a logic 0, followed by a programmable number of data bits (typically between 6 and 8). The receiver detects the start bit by sensing a falling edge (logic 1 to 0 transition) and then samples each subsequent bit at the center of its bit period, using the internally generated baud clock. The stop bit must always be logic 1.

UART’s simplicity and robustness make it a widely used solution for embedded systems and serial communication with peripheral devices [7].

Analog Protocol

The Arduino ATmega controllers used for the Arduino contain an onboard 6 channel analog-to-digital (A/D) converter. The converter has 10 bit resolution, returning integers from 0 to 1023. While the main function of the analog pins for most Arduino users is to read analog sensors, the analog pins also have all the functionality of general purpose input/output (GPIO) pins (the same as digital pins 0 - 13) [22].

In this context, it is suggested that the Arduino `analogRead` function work along with some source code that executes calibration function before starts gathering the actual data. This would increase the reliability of the readings with a small toll on the amount of data spent on measuring. The reason why mapping is required is that the Arduino can only return integers from 0 to 1023. Arduino analog pins compare the input voltage on each pin to a reference voltage, which in turn is the operating voltage 5V. So this practice interprets the measurement input from the sensor as a value within the range [0-1023].

$$\text{analogRead}() = \left(\frac{V_{\text{reading}}}{V_{\text{reference}}} \right) \times 1023 \quad (2.33)$$

Another practice that can also increase the reliability of the readings is the usage of some statistical means that would decrease the chances of chances of reading false data. Some exquisite practices are the average value, the median value and the average value after getting rid of the minimum and maximum values.

The aforementioned practices are not mandatory for this context. In addition to this they reduce severely the amount of data within the reading, which leads to smaller amount of data on frequency domain, after carrying out the Fast Fourier Transform. For this reason, raw data

that can go through some mapping-calibration is the ideal practice.

Chapter 3

System Engineering and Design

3.1 Introduction

As it has already been articulated, the objective of this diploma thesis is to study, design, and develop a complete measuring system that integrates a sensor, a microcomputer, and a microcontroller into a coherent and functional unit. The intention behind this effort is not only to demonstrate the feasibility of such a system, but also to provide a framework that can be applied to real-world industrial monitoring scenarios.

The system is specifically conceived for low-cost, real-time vibration monitoring of rotating machinery, a task of significant importance in predictive maintenance and condition monitoring. By capturing the vibrational behavior of machines during operation, valuable insights can be obtained about their mechanical health, potential faults, and overall performance. This contributes directly to reducing downtime, optimizing maintenance schedules, and extending the operational lifespan of equipment.

The chosen architecture relies on a modular set-up that combines hardware and software components of major importance. At its core lies a sensor element capable of detecting vibrations within a predetermined, application-relevant frequency range. The signals acquired are then transferred to a microcontroller, which is tasked with initial data handling, conditioning, and communication. Subsequently, the processed data is relayed to a microcomputer that ensures reliable storage in a structured database, while also providing the computational capacity required for further analysis and processing. In this manner, the system covers the complete cycle of measurement: from data acquisition, to secure storage, and finally to intelligent interpretation.

The following sections of this thesis provide a more detailed overview of the system requirements, addressing both functional requirements (such as measurement accuracy, sampling rates, and communication protocols) and non-functional requirements (such as cost efficiency, scalability, and reliability). Furthermore, a thorough elaboration on the system set-up is presented (see Figure 3.1), offering a comprehensive description of the interaction between hardware modules, data flow, and the software infrastructure that binds them together.

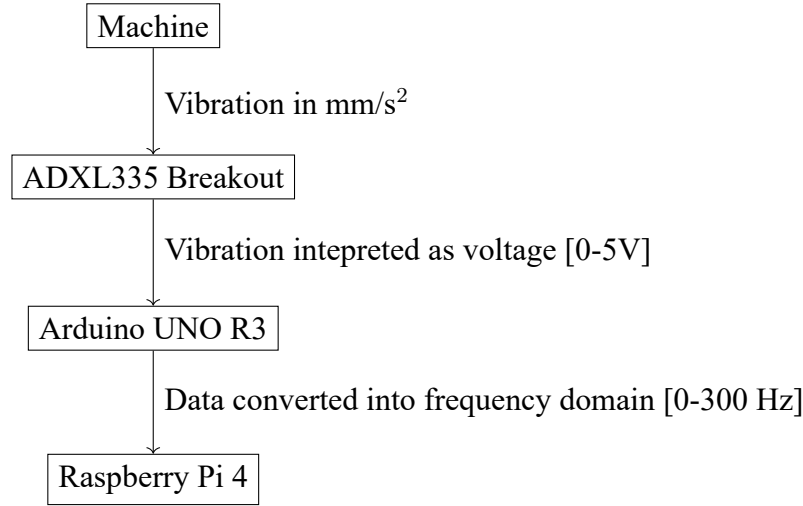


Figure 3.1: The concept layout.

3.2 Hardware Architecture

3.2.1 ADXL335 Accelerometer

The ADXL335 is a small, low-power, 3-axis accelerometer capable of measuring acceleration with a minimum full-scale range of $\pm 3g$. It is designed for motion- and tilt-sensing applications, making it suitable for mobile devices, gaming systems, disk drive protection, image stabilization, and sports and health devices. The device operates on a single supply voltage ranging from $1.8V$ to $3.6V$ and consumes only $350\mu A$ of current, making it ideal for power-sensitive applications [21].

The Adafruit breayout of this very sensor has a power supply at $5V$ which is cut off according to the needs of the sensor. The VCC takes up to $5V$ in and regulates it to $3.3V$ with an output pin. The analog outputs are ratiometric: that means that $0g$ measurement output is always at half of the $3.3V$ output ($1.65V$), $-3g$ is at $0v$ and $3g$ is at $3.3V$ with full scaling in between.

The aforementioned Adafruit ADXL335 accelerometer boasts a wide variety of amendments in comparison with the original stand-alone sensor. Most importantly, this breakout is an ideal choice when it comes to interfacing with 5V microcontrollers like Arduino. It is compatible with microcontrollers because there is pending connections on the breakout. Alternatively, a PCB must be designed so as to solder the small scale and hard to manage pins of the sensor itself.

Another significant feature of the breakouts is that manufacturing research has been conducted and this has lead to the optimal product development. This entails that these breakouts encompass decoupling capacitors, filters and power supply regulation that assures the pinout signals to be as accurate as possible.

It can measure the static acceleration of gravity in tilt-sensing applications, as well as dynamic acceleration resulting from motion, shock, or vibration.

A figure depicting the ADXL335 accelerometer is depicted on Figure 2.8 of Chapter 2. The ADXL335 is housed in a 16-lead LFCSP (Low Profile Chip Scale Package). Table 3.1 describes its key pin functions.

Table 3.1: ADXL335 Pin Functions [21]

Pin Number	Pin Name	Description
2	ST	Self-Test pin. When set to V_S , it generates an electrostatic force to test functionality.
3,5,6,7	COM	Common ground pins.
12	X_{OUT}	X-axis analog output.
10	Y_{OUT}	Y-axis analog output.
8	Z_{OUT}	Z-axis analog output.
14,15	V_S	Supply voltage pins (1.8 V to 3.6 V).
-	Exposed Pad	Not internally connected but should be soldered for mechanical integrity.

Table 3.2 summarizes the key specifications of the ADXL335 accelerometer [21].

All in all, the ADXL335 is widely used in applications requiring precise motion detection and tilt sensing. Its excellent temperature stability and low power consumption make it suitable for portable and battery-operated devices. The device's performance is further enhanced by its low noise density and adjustable bandwidth, allowing for customization based

Table 3.2: ADXL335 Key Specifications [21]

Parameter	Value
Measurement Range	$\pm 3\text{ g}$ (minimum), $\pm 3.6\text{ g}$ (maximum)
Sensitivity	300 mV/g (typical) at $V_S = 3\text{ V}$ (ratiometric)
Zero g Bias	1.5 V (typical) for X and Y axes, 1.5 V (typical) for Z axis
Noise Performance (X,Y axes)	$150\text{ }\mu\text{g}/\sqrt{\text{Hz}}$ (typical)
Noise Performance (Z axis)	$300\text{ }\mu\text{g}/\sqrt{\text{Hz}}$ (typical)
Bandwidth (X,Y)	Adjustable up to 1600 Hz via external capacitors
Bandwidth (Z)	Adjustable up to 550 Hz via external capacitors
Operating Temperature	-40°C to $+85^\circ\text{C}$
Shock Survival	$10,000\text{ g}$ (unpowered or powered)

on specific application requirements [21].

For further details, refer to the ADXL335 datasheet [21].

3.2.2 Arduino Uno

The Arduino UNO represents one of the most accessible and widely adopted micro-controller platforms in the maker community. Built around the reliable ATmega328P architecture, this development board strikes an optimal balance between capability and user-friendliness, serving equally well as an educational tool for novices and a rapid prototyping solution for experienced engineers. Its plug-and-play design eliminates complex setup procedures - users can immediately begin development by connecting via standard USB or utilizing various power input methods including the convenient barrel jack for 9V battery operation.

The specifications of Arduino are listed below [23]:

- **ATmega328P Microcontroller Unit** – Serving as the computational core, this component delivers:
 - Efficient 8-bit AVR RISC processing at 16 MHz clock speed
 - Substantial 32 kilobyte Flash memory for program storage
 - 2 kilobytes of volatile SRAM for dynamic memory operations
 - 1 kilobyte EEPROM for persistent data storage across power cycles

- **System Reliability Features:**

- Automatic Power-On Reset (POR) circuitry for stable initialization
- Brown-Out Detection (BOD) protection against voltage fluctuations

- **Integrated Hardware Peripherals:**

- Dual 8-bit and single 16-bit hardware timers with advanced control capabilities
- Full-duplex USART interface with sophisticated clock generation
- Flexible SPI and I2C serial communication controllers
- Precision analog comparator with configurable reference
- System watchdog for fail-safe operation
- Six-channel PWM generation for analog signal emulation
- Pin-change interrupt system for efficient event handling
- Seamless bridge between USB host and USART peripheral
- Automatic protocol conversion (USB packets ↔ serial frames)
- Built-in drivers for major operating systems

- **USB Interface Controller (ATmega16U2):**

- Dedicated 8-bit processor managing USB protocol
- 16 kilobyte reprogrammable Flash memory
- 512 bytes each of EEPROM and SRAM
- On-chip debugging through debugWIRE interface

- **Input/Output Configuration:**

- Fourteen programmable digital I/O lines (with six supporting PWM)
- Six-channel 10-bit analog-to-digital conversion
- Standard USB Type-B connector for programming and power
- In-circuit serial programming (ICSP) header
- Dedicated hardware reset circuit

- **Power Management System:**

- Wide operating voltage range (2.7-5.5V DC)
- Multiple power input options including regulated 5V USB power

A concise overview of the available pins is given in Table 3.3. For completeness, the full pinout information is provided in the appendix in Tables .1 and .2, which list all analog, digital, and power connections in detail. These complement the schematic representation shown in Figure 3.2.

Table 3.3: Summary of Arduino Pin Functions

Category	Typical Pins	Description
Power Supply	+3.3V, +5V, GND, VIN	Provides operating voltages and ground
Analog Inputs	A0–A5	6 analog channels, some also used for I ² C
Digital I/O	D0–D13	General-purpose digital pins, PWM on some
Communication	SDA, SCL, MOSI, MISO, SCK, SS	Interfaces for I ² C and SPI
Special	Reset, IOREF, AREF	Reset, voltage reference, and analog reference

An exquisite trait of Arduino boards, is that these devices boast a built-in set up of pull-up and pull-down resistors, which is extremely helpful when it comes to implementing projects making use of some sensors.

Moreover, the Arduino UNO cleverly bridges modern USB communication with traditional serial (UART) communication through a two-chip design. The main ATmega328P microcontroller handles the actual programming and serial communication using simple UART protocol, while a secondary ATmega16U2 chip acts as a translator, converting between the computer's USB signals and the microcontroller's UART signals. This translation happens automatically in both directions - when you upload code, the USB chip converts your computer's USB commands into UART signals the main chip understands, and when the Arduino sends serial data back, the USB chip converts it back to USB for your computer. The system creates a virtual COM port on your computer, making the Arduino appear as a simple

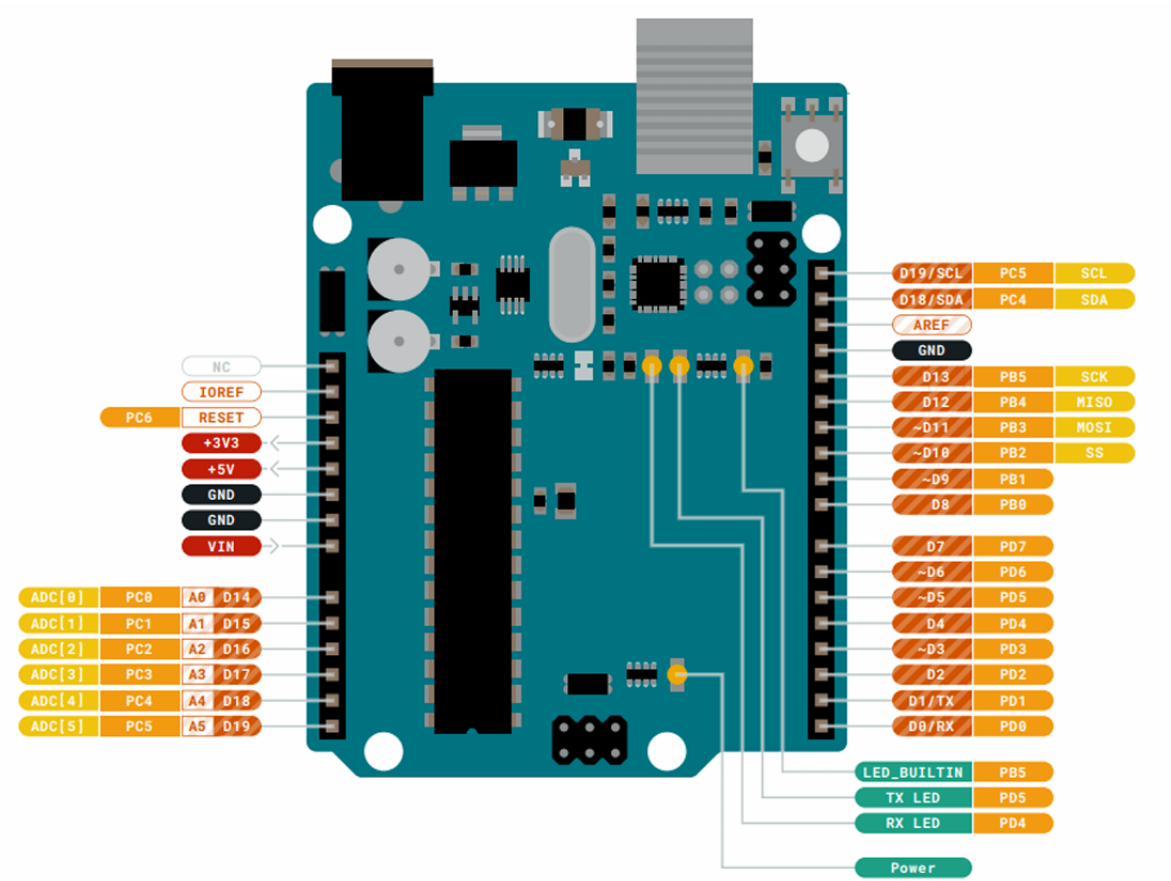


Figure 3.2: Arduino schematic pinout (refer to Tables .1 and .2 for details).

serial device even though it’s actually communicating via USB. This elegant solution provides plug-and-play USB connectivity while maintaining compatibility with traditional serial communication, all without requiring any additional components or complex setup from the user. The dual-chip approach also keeps the main processor free from USB protocol handling, allowing it to focus on running your programs efficiently.

The Arduino UNO’s serial communication typically operates at standard baud rates ranging from 300 to 115200 bits per second, with 9600 baud being the most commonly used default speed for basic communication. The USB-UART bridge supports higher speeds up to 2 Mbps in theory, but in practice, most applications reliably use 115200 baud or lower. The actual stable maximum baud rate depends on several factors, including the clock accuracy (the UNO’s 16MHz crystal provides $\pm 0.3\%$ stability), the quality of USB connections, and the timing requirements of the specific application. For most beginner projects and serial monitoring, 9600 or 115200 baud provide the best balance of speed and reliability, while advanced applications can push higher speeds if properly implemented. The baud rate must match on

both the Arduino and the receiving device for successful communication.

There is also a tremendous amount of source code that is adaptable to different projects. All these sketches can be easily downloaded, stored and modified (or just even tweak some parameters) so as to fit it in each project.

All in all, the UNO is the most popular and well-documented board in the Arduino ecosystem, making it perfect for learning electronics and coding. Its processor ATmega328P chip, in most of the cases is not welded and this allows easy replacement procedure. In addition Arduino UNO is robust, flexible, and beginner-friendly platform for a wide variety of projects. Most importantly it supports easy and robust communication between Raspberry Pi and Arduino.

For further details, refer to the Arduino UNO R3 datasheet [23].

3.2.3 Raspberry Pi 4

The Raspberry Pi is a series of credit-card sized single-board computers developed by the Raspberry Pi Foundation. Designed originally for education, these low-cost computers have evolved into versatile platforms for computing and digital making [8].

There is a plethora of current Raspberry Pi boards (e.g. Raspberry Pi 4 and Raspberry Pi 5). They are all extremely powerful and can cover the needs of highly demanding projects, in terms of memory and computational power [8]. The Raspberry Pi's complete technical specifications are systematically organized in Tables 3.4, 3.5, and 3.6, yielding detailed hardware and software characteristics.

Table 3.4: Key Specifications of Raspberry Pi

Component	Specification
Processor	Broadcom BCM2712 (Pi 5) or BCM2711 (Pi 4) ARM Cortex
Memory	4GB LPDDR4X RAM
Storage	MicroSD card (64GB recommended)
Power Supply	5V USB-C (3A recommended)
Connectivity	Dual-band 802.11ac Wi-Fi, Gigabit Ethernet
Video Output	Dual micro HDMI (up to 4K resolution)
GPIO	40-pin header

Table 3.5: Raspberry Pi OS Features

Feature	Description
OS Type	Debian-based Linux distribution
Desktop	Complete desktop environment
Development	Python tools pre-installed
Hardware Access	GPIO control libraries

Table 3.6: Supported Communication Protocols

Protocol	Description
Digital I/O	General purpose input/output pins
SPI (Serial Peripheral Interface)	High-speed serial communication
I ² C (Inter-Integrated Circuit)	Two-wire serial bus
UART	Asynchronous serial communication

A clear representation of the Raspberry Pi 4 hardware is depicted on the Figure 3.3. Raspberry Pi boasts a wide variety of input and output pins, resembling the capacity of Arduino. It is also programmable, having a source code back-up for multiple hardware options (sensors). The most frequently used coding language for this context is PYTHON.

The Raspberry Pi 4 Model B offers comprehensive connectivity options as detailed in Table 3.7. The board features dual display outputs through micro-HDMI ports capable of driving 4K resolution simultaneously. For peripheral connectivity, it provides a balanced mix of USB 3.0 and 2.0 ports, with the USB-C power input supporting higher current requirements. The onboard wireless interfaces deliver dual-band Wi-Fi and Bluetooth 5.0 for modern wireless applications.

Table 3.8 details the 40-pin GPIO header configuration, which maintains backward compatibility with previous Raspberry Pi models. The header provides multiple voltage rails (3.3V and 5V) along with dedicated ground pins. The GPIO header serves as the primary interface for hardware projects, supporting everything from simple LED controls to complex sensor networks and custom communication protocols.

For further details, refer to the Raspberry Pi 4 Model B datasheet [8].

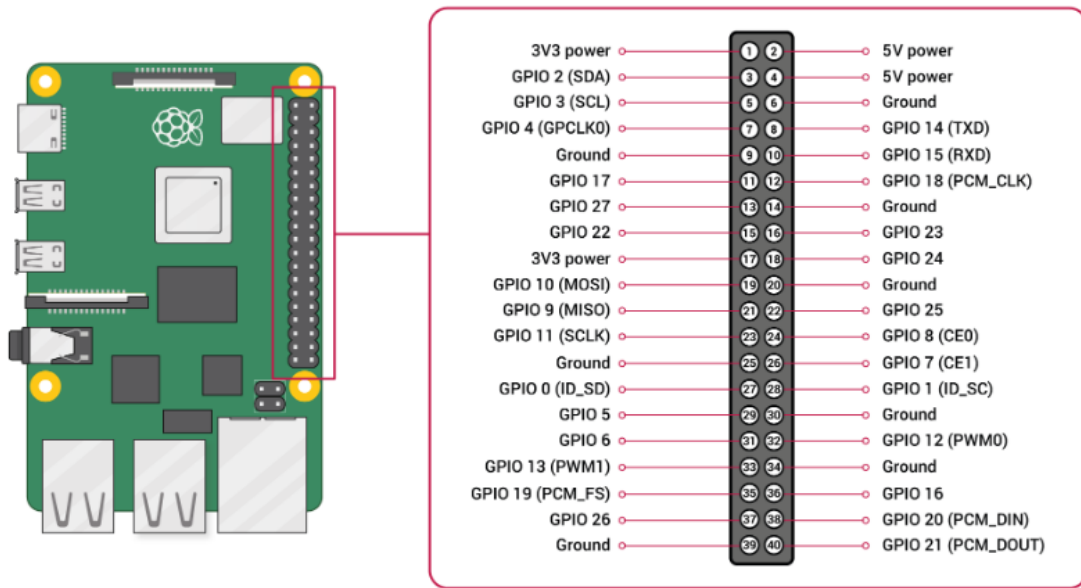


Figure 3.3: 40-PIN GPIO of Raspberry Pi 4 (source [8]).

3.2.4 The whole set-up

Following the presentation of the different ports of both the Raspberry Pi 4 and the Arduino UNO R3, it is important to also describe the interconnection between these two devices and the way in which data flows through the system. The accelerometer module (Adafruit ADXL335 breakout) generates analog voltage signals corresponding to the acceleration along the three axes (X, Y, and Z). These signals are connected to the analog input pins of the Arduino UNO, which then uses its built-in Analog-to-Digital Converter (ADC) to digitize the incoming data.

After conversion, the Arduino transmits the measurements via serial communication to the Raspberry Pi 4. On the Raspberry Pi side, a Python script is used to set up the serial connection, send requests for new measurements, and receive the acceleration values. Specifically, the Raspberry Pi sends a command string to the Arduino, which responds with a comma-separated line containing the three acceleration components. The incoming data stream is subsequently retrieved as a text line, converted into a human-readable string, and then parsed into numerical values corresponding to the three acceleration axes. Finally the data is split into three floating-point numbers representing the X, Y, and Z axes.

The baud rate of the serial communication, which must be pre-set identically on both the Arduino and the Raspberry Pi, plays a crucial role in ensuring reliable data transfer. The baud

Table 3.7: Raspberry Pi 4 Model B Ports and Connectors

Port Name	Type	Quantity	Specifications
Primary External Ports			
USB 3.0	Type-A	2	5Gbps, BC1.2 charging support
USB 2.0	Type-A	2	480Mbps
Ethernet	RJ45	1	Gigabit (Realtek RTL8111G)
HDMI	micro-HDMI	2	HDMI 2.0 (up to 4K@60Hz)
Power	USB-C	1	5V/3A (minimum)
Audio	3.5mm jack	1	Stereo output + composite video
GPIO Header			
GPIO	40-pin	1	2×20 pin header, 0.1” spacing
Other Connectors			
CSI	15-pin FPC	1	Camera Serial Interface
DSI	15-pin FPC	1	Display Serial Interface
MicroSD	Slot	1	UHS-I compatible
PoE	4-pin header	1	802.3af support (requires HAT)
Wireless Interfaces			
Wi-Fi	Antenna	1	2.4GHz/5GHz 802.11ac
Bluetooth	Antenna	1	BT 5.0, BLE

rate defines the speed at which bits are transmitted per second, and therefore determines how much information can be sent within a given time window. If the chosen baud rate is too low compared to the frequency and volume of data produced by the Arduino, the serial buffer may overflow, leading to delays or data loss. On the other hand, selecting a sufficiently high baud rate allows the system to transmit each line of acceleration data quickly enough to keep up with the sensor output, ensuring that all measurements are captured in real time. For example, at 9600 bps only around 960 bytes per second can be transmitted, which may not be adequate for high-frequency sampling, whereas at 115200 bps more than 11,000 bytes per second can be handled, providing ample capacity for continuous vibration monitoring. In this case, both devices can work well with a baud rate at 115200 bps.

In summary, the flow of information can be described as follows: the ADXL335 breakout produces analog signals sending them to the Arduino UNO samples and digitizes them. Af-

Table 3.8: GPIO Pinout Details (An extended version of this table is on Appendix - extended version of the following table is the Table .3 (see Chapter 5.2).

Pin #	Function	Alt Function	Voltage	Notes
1	3.3V	-	3.3V	50mA max
2	5V	-	5V	-
3	GPIO2	SDA1	3.3V	I ² C
4	5V	-	5V	-
5	GPIO3	SCL1	3.3V	I ² C
6	GND	-	-	-
7	GPIO4	-	3.3V	-
...	...	...	...	...
37	GPIO26	-	3.3V	-
38	GPIO20	MOSI1	3.3V	SPI
39	GND	-	-	-
40	GPIO21	MISO1	3.3V	SPI

terwards the data is transmitted through the serial interface to the Raspberry Pi which in turn requests, receives, and parses the acceleration values for further storage and analysis. This interaction ensures a continuous and reliable stream of vibration data, which is schematically represented in Figure 3.4.

3.3 System Requirements

3.3.1 Functional Requirements

The most commonly rotating speed of machinery goes up to 1500 [rpm] corresponds to 25 [Hz]. This relationship is defined by the following equation:

$$f \text{ [Hz]} = \frac{r \text{ [RPM]}}{60} \quad (3.1)$$

The aforementioned relationship entails that the rotational speed of a machine can be interpreted in frequency, which in turn is of great help when related to Fourier Transform applications.

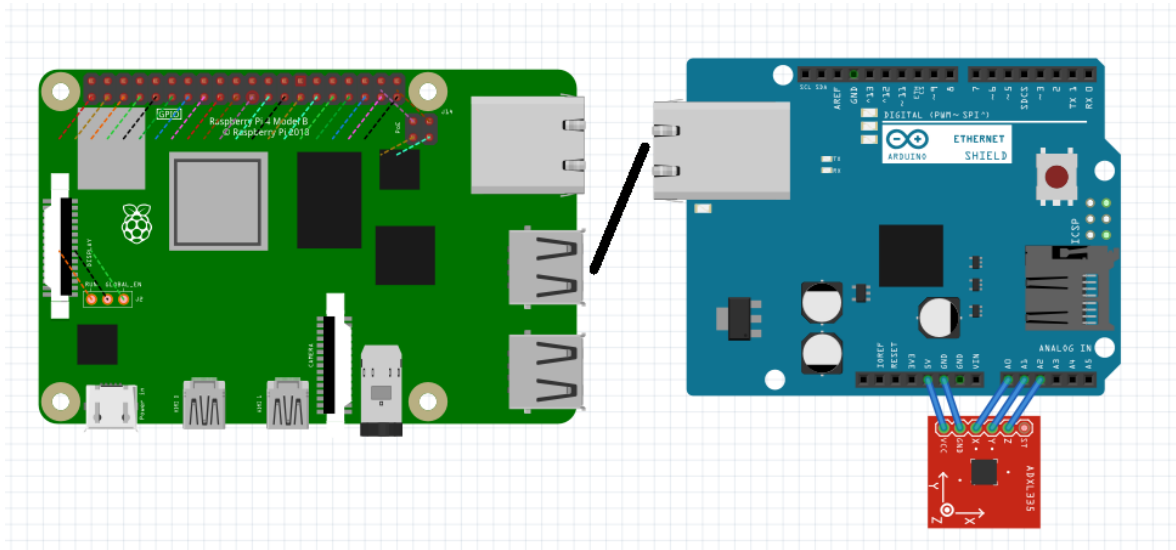


Figure 3.4: Hardware architecture with signal flow.

All in all, the functional requirements are highlighted hereunder:

- **Frequency Range:** 0–300 Hz (covers common machine faults like imbalance, misalignment).
- **Real-Time Processing:** On-the-fly FFT computation for immediate fault detection.
- **Scalability:** Modular design for multi-sensor deployments.

3.3.2 Non-Functional Requirements

- **Cost:** Minimize BOM cost (justifies Arduino + RPi4 over industrial PLCs).
- **Power:** Optimize for continuous operation (e.g., no active cooling).
- **Accuracy:** ADXL335's $\pm 3g$ range suffices for industrial vibrations (cite datasheet).

3.4 Software Architecture and Data Flow

3.4.1 Signal Processing Pipeline

The Arduino samples data at 600Hz, which is extremely high and constitutes a sufficient sample rate for this project. Additionally, another consideration is receiving data from the ADXL335 sensor breakout, which could potentially make this procedure slower. The highest data volume over a predefined time span achieved for this project is 300Hz.

In essence, a batch of data is received through this procedure by a Raspberry Pi 4, which in turn processes and analyzes these samples.

3.4.2 Scalability Considerations

- **Multi-Threading:** RPi4 handles concurrent sensor inputs (e.g., 4x Arduino nodes).
- **Data Compression:** FFT bin reduction (0–300Hz only) minimizes cloud costs.

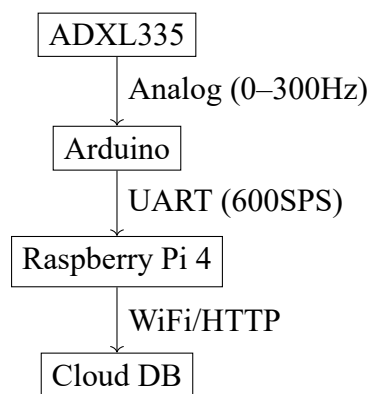


Figure 3.5: End-to-end data flow with protocols

3.4.3 Arduino sketch

Elaboration on (including the whole code):

- Description of sketch and cpp files
- elaborate on how it shares data via UART
- Analyse the possibility of translating Voltage into g or m/s²

3.4.4 Raspberry Pi coding

Raspberry Pi 4:

- Runs Python-based FFT (e.g., NumPy) and SQLite for local storage.
- WiFi/Bluetooth enables cloud integration (e.g., AWS IoT, InfluxDB).

Elaboration on (including some parts of the code):

- Description of the whole program
- how it posts data via HTTP
- Data stored on database , and database setup

3.4.5 Vercel coding

Elaboration on (including some parts of the code):

- page file that prints data
- how it is coupled with supabase

Chapter 4

Results

4.1 Intro

- depict data dashboard
- analyse alarm cases
- database - stored data

Chapter 5

Conclusions

Εδώ εξηγούμε ότι θα συνοψίσουμε την μελέτη που εκπονήθηκε στα πλαίσια της διπλωματικής.

- summarize how it decides if a bearing is defective
- how it collects data
- how data is analysed and at what stages it is analysed

5.1 Σύνοψη και συμπεράσματα

Εδώ συνοψίζουμε τα αποτελέσματα της διπλωματικής και περιγράφουμε τα συμπεράσματα που προέκυψαν, αρνητικά και θετικά. Επιβεβαιώνουμε την συνεισφορά της διπλωματικής στα προβλήματα που αναφέραμε στην εισαγωγή.

5.2 Μελλοντικές επεκτάσεις

Εδώ δίνουμε ιδέες για επέκταση της διπλωματικής.

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APPENDICES

Appendix

Τίτλος Παραρτήματος

1 Tables

Table .1: Analog Pins (JANALOG)

Pin	Function	Type	Description
1	NC	NC	Not connected
2	IOREF	IOREF	Reference for digital logic V (connected to 5V)
3	Reset	Reset	Reset
4	+3V3	Power	+3.3V Power Rail
5	+5V	Power	+5V Power Rail
6	GND	Power	Ground
7	GND	Power	Ground
8	VIN	Power	Voltage Input
9	A0	Analog/GPIO	Analog input 0 / GPIO
10	A1	Analog/GPIO	Analog input 1 / GPIO
11	A2	Analog/GPIO	Analog input 2 / GPIO
12	A3	Analog/GPIO	Analog input 3 / GPIO
13	A4/SDA	Analog input/I2C	Analog input 4 / I2C Data line
14	A5/SCL	Analog input/I2C	Analog input 5 / I2C Clock line

Table .2: Digital Pins (JDIGITAL)

Pin	Function	Type	Description
1	D0	Digital/GPIO	Digital pin 0 / GPIO
2	D1	Digital/GPIO	Digital pin 1 / GPIO
3	D2	Digital/GPIO	Digital pin 2 / GPIO
4	D3	Digital/GPIO	Digital pin 3 / GPIO
5	D4	Digital/GPIO	Digital pin 4 / GPIO
6	D5	Digital/GPIO	Digital pin 5 / GPIO
7	D6	Digital/GPIO	Digital pin 6 / GPIO
8	D7	Digital/GPIO	Digital pin 7 / GPIO
9	D8	Digital/GPIO	Digital pin 8 / GPIO
10	D9	Digital/GPIO	Digital pin 9 / GPIO
11	SS	Digital	SPI Chip Select
12	MOSI	Digital	SPI Main Out Secondary In
13	MISO	Digital	SPI Main In Secondary Out
14	SCK	Digital	SPI Serial Clock Output
15	GND	Power	Ground
16	AREF	Digital	Analog Reference Voltage
17	A4/SD4	Digital	Analog input 4 / I2C Data line (duplicated)
18	A5/SD5	Digital	Analog input 5 / I2C Clock line (duplicated)

Table .3: Complete GPIO Pinout for Raspberry Pi 4 Model B (Part 1/2)

Pin #	Function	Alt Function	Voltage	Notes
1	3.3V	—	3.3V	50mA max
2	5V	—	5V	Power supply
3	GPIO2	SDA1	3.3V	I ² C Data
4	5V	—	5V	Power supply
5	GPIO3	SCL1	3.3V	I ² C Clock
6	GND	—	—	Ground
7	GPIO4	GPCLK0	3.3V	General purpose
8	GPIO14	TXD0	3.3V	UART Transmit
9	GND	—	—	Ground
10	GPIO15	RXD0	3.3V	UART Receive
11	GPIO17	—	3.3V	General purpose
12	GPIO18	PCM_CLK	3.3V	PWM0
13	GPIO27	—	3.3V	General purpose
14	GND	—	—	Ground
15	GPIO22	—	3.3V	General purpose

Table .4: Complete GPIO Pinout for Raspberry Pi 4 Model B (Part 2/2) (Continued from Table .3)

Pin #	Function	Alt Function	Voltage	Notes
16	GPIO23	–	3.3V	General purpose
17	3.3V	–	3.3V	50mA max
18	GPIO24	–	3.3V	General purpose
19	GPIO10	MOSI0	3.3V	SPI0
20	GND	–	–	Ground
21	GPIO9	MISO0	3.3V	SPI0
22	GPIO25	-	3.3V	General purpose
23	GPIO11	SCLK0	3.3V	SPI0
24	GPIO8	CE0	3.3V	SPI0 Chip Enable 0
25	GND	-	-	-
26	GPIO7	CE1	3.3V	SPI0 Chip Enable 1
27	ID_SD	-	-	EEPROM ID
28	ID_SC	-	-	EEPROM Clock
29	GPIO5	-	3.3V	General purpose
30	GND	-	-	-
31	GPIO6	-	3.3V	General purpose
32	GPIO12	PWM0	3.3V	PWM channel 0
33	GPIO13	PWM1	3.3V	PWM channel 1
34	GND	-	-	-
35	GPIO19	PCM_FS	3.3V	PWM/MISO1
36	GPIO16	-	3.3V	General purpose
37	GPIO26	-	3.3V	General purpose
38	GPIO20	MOSI1	3.3V	SPI1
39	GND	-	-	-
40	GPIO21	MISO1	3.3V	SPI1

1.1 Αλγόριθμοι

Ακολουθεί ο Αλγόριθμος 1, ο οποίος είναι μορφοποιημένος με τα πακέτα `algorithm` και `algorithmic`.

Algorithm 1 Probabilistic $k\theta NN$ Monitoring

```

1: Procedure VerifyCandidate (focal query point  $q$ , threshold  $\theta$ , object  $o$ , list of auxiliary objects  $P$ ,
   distance  $kMAXDIST$ )
2: if  $\Phi(o, kMAXDIST) \geq \theta$  and  $L_2(q, o) \leq L_2(q, P.top())$  then
3:    $P.pop()$ ;    //Replace the most extreme element in  $P$ , since candidate  $o$  ...
4:    $P.push(o)$ ;  //... has enough probability and has its mean closer to focal  $q$ 
5: end if
6: End Procedure
  
```

1.2 Θεωρήματα, Πορίσματα, Ορισμοί, κλπ.

Ακολουθεί παράδειγμα θεωρήματος από την ιστοσελίδα https://www.overleaf.com/learn/latex/Theorems_and_proofs

Theorem 5.1. *Let f be a function whose derivative exists in every point, then f is a continuous function.*